# Model Memory vs Retrieval

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

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TEACHER GUIDE · 1 OF 9

## Model Memory vs Retrieval

> Prepare Model Memory vs Retrieval: check the target, prerequisites, resources and adaptable teaching schedule.

### Teach the learning cycle

Primary target: Explain the **mechanism** and terminology of **retrieval** and **index**ing

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

### Retrieval answer guidance

### Verify Numbers and Dates

Without opening the old lesson, explain one key idea from Verify Numbers and Dates. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of claim support and citation evaluation

Reference example, not the only acceptable wording: Teacher explanation: Citation evaluation checks source existence, exact passage support and coverage of answer claims. A real citation can still be irrelevant.

For this classroom rubric count a claim as supported only when its cited passage entails it. Separately count correct citation links. Use the number of factual claims as the support denominator.

### Hallucination Failure Types

Without opening the old lesson, explain one key idea from Hallucination Failure Types. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of generation failures and verification

Reference example, not the only acceptable wording: Teacher explanation: Generated answers may invent facts, misread quantities or cite irrelevant passages.

Verification checks each claim against **evidence** rather than accepting fluent wording. Split an answer into atomic claims. For each, record the exact source passage, support status and any needed calculation.

Mark unsupported and contradicted claims separately; do not repair them by inventing sources.

### Visual Composition

Without opening the old lesson, explain one key idea from Visual Composition. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of image generation and provenance

Reference example, not the only acceptable wording: Teacher explanation: An image generator constructs an **output** from learned patterns and a specification. A generated picture is not a record that the depicted event occurred.

Keep a provenance record containing specification, tool/version when known, generation date, edits, permissions and intended use. Label fictional imagery.

Inspect visual requirements and avoid inferring identity or event authenticity from appearance.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

### What the learner must demonstrate

A model states the library opens at 8 from learned patterns. A supplied current notice says 9. Explain the stronger answer and one remaining check.

### Concept to explain

Learned **parameters** are not a searchable source library.

| Term | Meaning |
| --- | --- |
| parameters | Values within a model adjusted during training; distinguish them from settings chosen by a developer. |
| retrieval | Finding and returning existing information from a source or collection. |
| grounding | Connecting an answer or action to specified evidence or observations. |

| Term | Meaning |
| --- | --- |
| abstain | Decline to make a prediction when the available evidence does not meet the decision rule. |

### Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

### Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

### Preparation for the connected explanation

Two 45-minute sessions. **Retrieval** returns stored records selected for a query. An **index** is a searchable representation of a versioned **corpus**; generated wording is a later operation.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

### Readiness answer

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short **retrieval** activity before this chapter.

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

**End goal for the pictorial case**

By the end, I can explain model memory vs **retrieval** using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

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## Explain the mechanism

> Model Model Memory vs Retrieval using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

**Input → Process → Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Model memory may produce a plausible general explanation.

2. A current date or exact quotation requires **retrieval**.

3. Arithmetic should be calculated, not guessed from patterns.

4. When source or tool is unavailable, state the limit and **abstain**.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

**Concrete teaching case**

### The teaching case

One assistant answers without search; another retrieves a handbook paragraph.

### The teaching case

Compare **evidence** trails.

**Exact explanation to model**

### Why the result follows

The first uses learned patterns without an inspected source. The second supplies a passage that can be checked. **Retrieval** still does not guarantee correct use.

### Explain the added worked example

**Retrieval** returns stored records selected for a query. An **index** is a searchable representation of a versioned **corpus**; generated wording is a later operation.

## Explain the added worked example · 1

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Assign stable document IDs, versions and permitted access before **index**ing. Retrieve only authorised records. Preserve source references and define behaviour when a document is missing, outdated or access-restricted.

## Explain the added worked example · 2

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

### Key terms

| Term | Meaning |
|---|---|
| Corpus | The collection of records available for a task |
| Index | A searchable representation of that collection |
| Freshness | Whether a record is current for the requested time |

### Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

**Index** v1 contains notice N1 for room A. **Corpus** v2 supersedes it with N2 for room B, but the query still returns N1. What can be reported?

The stale **index** returns old **evidence**. It cannot establish current room approval; refresh against authorised **corpus** v2 or report unresolved **freshness**.

### Model the reasoning aloud · 3

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

| Thinking move | Teacher prompt |
|---|---|
| Plan | What is given? What is the question? Which procedure fits these facts? |
| Do | Point to each input and intermediate step in the worked trace. Name the rule that allows the next step. |
| Check | Does the output answer the question? Which assumption or limitation prevents a larger claim? |

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

### A pictorial worked case: Model Memory vs Retrieval

The library assistant remembers an old opening time, but a current notice can be retrieved.

### The facts we will use

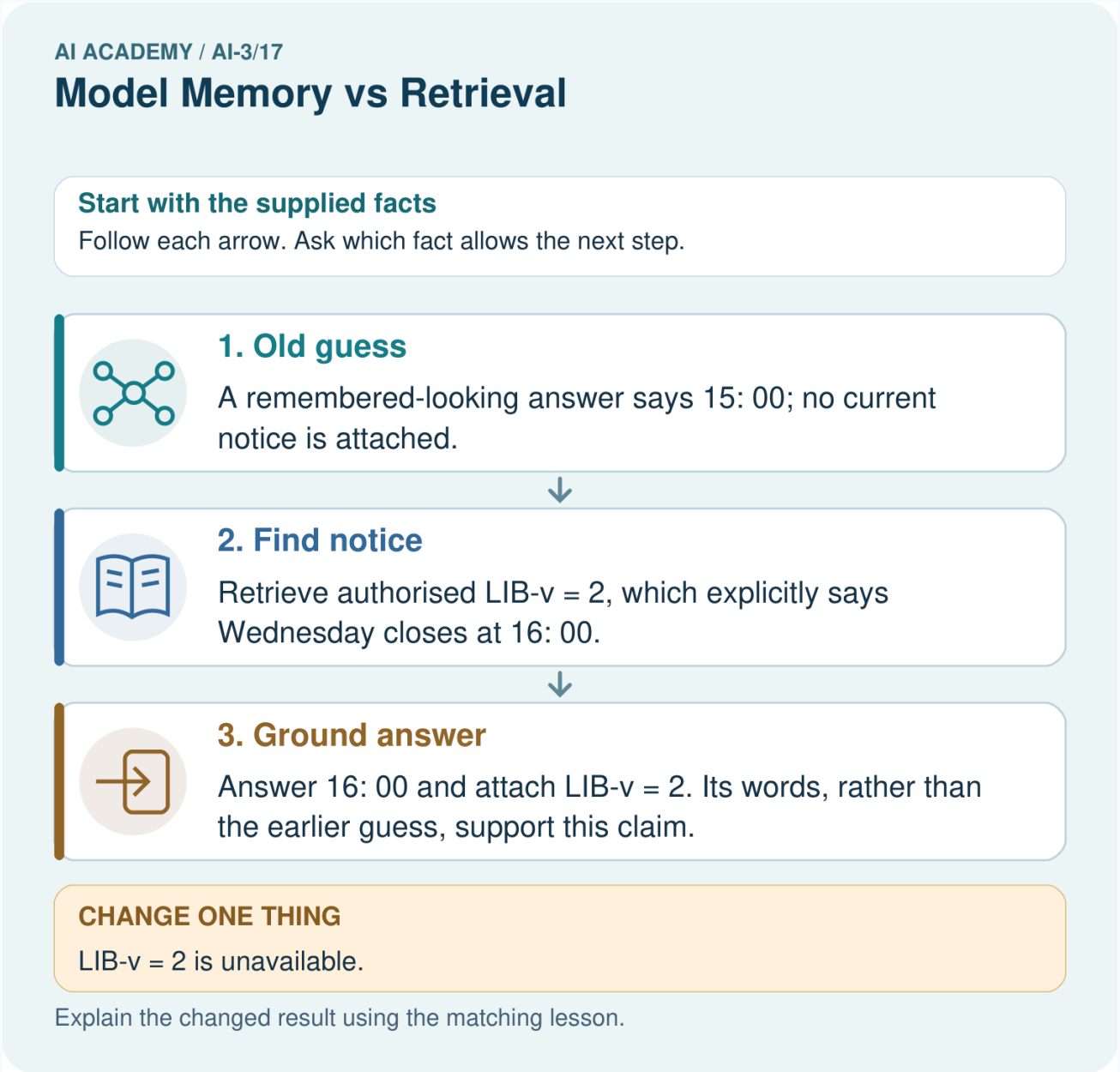
#### Supplied facts

Approved notice LIB-v = 2 says Wednesday closes at 16: 00; the model previously produced

15: 00. The question asks the current Wednesday closing time.

### Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.




Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Old guess | A remembered-looking answer says 15: 00; no current notice is attached. |
| 2. Find notice | Retrieve authorised LIB-v = 2, which explicitly says Wednesday closes at 16: 00. |
| 3. Ground answer | Answer 16: 00 and attach LIB-v = 2. Its words, rather than the earlier guess, support this claim. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

### Why the result follows

Read the picture in order. Answer 16: 00 and attach LIB-v = 2. Its words, rather than the earlier guess, support this claim. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

### What this example does not establish

#### Keep the stated limit

A retrieved document can itself be wrong or outdated; source approval, date and scope still need checking.

### Change one thing in the pictured case

#### Change one thing

Change: LIB-v = 2 is unavailable.

#### Predicted result

**Prediction**: Report that the current time cannot be verified.

#### Why the result follows

Reason: The unsupported old guess does not become current **evidence** when **retrieval** fails.

### Ask while showing the picture

## Ask while showing the picture · 4

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

# Demonstrate and guide practice

> Demonstrate Model Memory vs Retrieval, reduce support deliberately and check individual reasoning before discussing answers.

**Board sequence**

### 1. Put the facts on the board

Write the given facts: One assistant answers without search; another retrieves a handbook paragraph.

### 2. State the decision to explain

Write the question to explain: Compare **evidence** trails.

### 3. Reveal the worked reasoning

Reveal after **prediction**s: The first uses learned patterns without an inspected source. The second supplies a passage that can be checked. **Retrieval** still does not guarantee correct use.

### Hold back the independent task

Keep for an independent attempt: A model recalls last year's opening hours; a retrieved current notice gives different hours. How should the answer use this **evidence**?

### Classroom dialogue

#### Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

#### Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

### Model these guided questions

Guided: The stale **index** returns old **evidence**. It cannot establish current room approval; refresh against authorised **corpus** v2 or report unresolved **freshness**.

Qualification: High **retrieval** similarity does not override permissions or make an obsolete source current.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

### Case-specific teaching sequence

| Stage | Teacher action |
| --- | --- |
| Session 1: 0–7 minutes | Retrieve the prerequisite; identify the input, procedure and evidence source. |
| 7–18 minutes | Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation. |

| Stage | Teacher action |
| --- | --- |
| 18–32 minutes | Trace the declared procedure and worked case. Complete W2, recording intermediate steps. |
| 32–45 minutes | Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations. |
| Session 2: 0–8 minutes | Retrieve the worked reasoning and collect the guided response before discussing it. |
| 8–20 minutes | Compare a proposed boundary variant with the original case; identify what remains fixed. |
| 20–35 minutes | Collect W4 independently. Require the procedure trace and the bounded conclusion. |
| 35–45 minutes | Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts. |

### Reduce help deliberately

#### Completion practice

Completion task: **Index** v1 contains notice N1 for room A. **Corpus** v2 supersedes it with N2 for room B, but the query still returns N1. What can be reported? Show the intermediate steps and give the conclusion with its scope.

#### Reasoning to listen for

Teacher check: Teacher explanation: The stale **index** returns old **evidence**. It cannot establish current room approval; refresh against authorised **corpus** v2 or report unresolved **freshness**.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

### Individual reasoning check

Use the worked case for **Retrieval** and **index**ing. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> #### Expected repair
>
> Expected correction: High **retrieval** similarity does not override permissions or make an obsolete source current.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

### Pictorial practice question

Why can the assistant not certify 15: 00 when the notice is missing? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

### Reasoned teacher answer

15: 00 came from an unsupported earlier **output**. Missing current **evidence** requires an uncertainty statement, not substitution. Changed-case check: Report that the current time cannot be verified.

The unsupported old guess does not become current **evidence** when **retrieval** fails.

Changed-case reasoning: Report that the current time cannot be verified. The unsupported old guess does not become current **evidence** when **retrieval** fails.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

# Run the weekly evidence lab

> Run the Model Memory vs Retrieval investigation and use the first unsupported step to choose a teaching response.

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Route twelve questions to model, **retrieval**, calculator, or **abstain**; defend the **evidence** requirement for each.

> What did the hardest test teach you that the normal test did not?

### Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

| Record | What to capture |
| --- | --- |
| Input or evidence | Exact data, instruction or claim |
| Expected result | Answer or safe fallback declared before the run |
| Observed result | Actual result, including errors |
| Explanation | Rule, calculation or evidence explaining the outcome |
| Next test | One justified change and a new prediction |

### Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

### Address these additional misconceptions

The worked result establishes every future case: High **retrieval** similarity does not override permissions or make an obsolete source current.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

### Match the support to the first error

| What the response shows | Next teaching action |
| --- | --- |
| Missing or misread facts | Read the case aloud, mark supplied versus unknown facts, and let the learner restate it. |
| Wrong procedure | Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation. |
| Arithmetic or implementation error | Trace intermediate values or lines; preserve the first mismatch and retry that step. |
| Correct result without a reason | Ask for a trace and a changed case before marking independent understanding. |
| An unsupported wider claim | Ask which supplied evidence supports the claim and which further observation would be needed. |

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

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## Assess and explain the answers

> Use the Model Memory vs Retrieval questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

### Worked case answer

**Teacher key · Worked case answer**

The first uses learned patterns without an inspected source. The second supplies a passage that can be checked. **Retrieval** still does not guarantee correct use.

### Independent question

**Independent question · withhold the key**

A model recalls last year's opening hours; a retrieved current notice gives different hours. How should the answer use this **evidence**?

### Expected independent answer

**Teacher key · Expected independent answer**

Use the relevant current notice, cite it and identify its date. Model recall is not a substitute for checking information that changes.

### What counts as evidence

| Criterion | 0 points | 1 point | 2 points |
|---|---|---|---|
| Result | Incorrect or absent | Partly correct | Correct with necessary conditions |
| Reasoning | No supporting trace | Some steps | Clear mechanism, calculation or source support |
| Limit | Overclaims certainty | Vague limitation | Relevant uncertainty or missing fact |
| Transfer | Copies without adapting | Adapts with prompts | Solves a changed case independently |

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

### Feedback

Ask the learner to find the **evidence** that challenges this claim: “A model's confident memory always outranks a current primary source.” Keep the first attempt and an uncoached retry separate.

### Original transfer answer

> #### Teacher key · Original transfer answer
>
> Answer 9 with the notice as support, after confirming the branch and date. Learned patterns are not an inspected current record.
>
> **Retrieval** helps provide **evidence** but the system can still retrieve or interpret the wrong passage.

## Explanations for the additional workbook examples

### W1 Explain the mechanism

#### Workbook question · W1

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

#### Teacher reasoning · W1

Teacher explanation: **Retrieval** returns stored records selected for a query. An **index** is a searchable representation of a versioned **corpus**; generated wording is a later operation.

Assign stable document IDs, versions and permitted access before **index**ing. Retrieve only authorised records. Preserve source references and define behaviour when a document is missing, outdated or access-restricted.

### W2 Trace the worked case

#### Workbook question · W2

**Index** v1 contains notice N1 for room A. **Corpus** v2 supersedes it with N2 for room B, but the query still returns N1. What can be reported? Show the intermediate steps and give the conclusion with its scope.

#### Teacher reasoning · W2

Teacher explanation: The stale **index** returns old **evidence**. It cannot establish current room approval; refresh against authorised **corpus** v2 or report unresolved **freshness**.

### W3 Audit the limit

#### Workbook question · W3

A report omits this qualification: High **retrieval** similarity does not override permissions or make an obsolete source current. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

#### Teacher reasoning · W3

Teacher explanation: High **retrieval** similarity does not override permissions or make an obsolete source current.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

### W4 Apply a new case independently

#### Workbook question · W4

**Corpus** has public P, teacher-only T and withdrawn W. A student search returns all three. Identify the filtering defect and whether hiding T after generation is sufficient.

#### Teacher reasoning · W4

Teacher explanation: Filter authorisation and active status before **evidence** reaches the student generation path: P only. Hiding a final citation does not undo exposed teacher content.

### W5 Design a revealing follow-up

#### Workbook question · W5

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

#### Teacher reasoning · W5

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. High **retrieval** similarity does not override permissions or make an obsolete source current.

The new case should test this particular **mechanism** rather than merely restate the original result.

### Assess independent transfer

#### Independent question · withhold the key

Selected case: **Corpus** has public P, teacher-only T and withdrawn W. A student search returns all three. Identify the filtering defect and whether hiding T after generation is sufficient.

### Selected case answer

#### Teacher key · Selected case answer

Teacher explanation: Filter authorisation and active status before **evidence** reaches the student generation path: P only. Hiding a final citation does not undo exposed teacher content.

| Dimension | Evidence required |
|---|---|
| Explanation | Names the applicable idea or procedure and ties it to the supplied facts. |
| Trace and result | Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution. |
| Limits | States the assumptions, missing evidence or conditions under which the result would change. |

#### Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

### Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

## Connect and extend learning

> Connect Model Memory vs Retrieval to Verified Study Buddy, adapt support and plan a later changed-case check.

### Verified Study Buddy

The Study Buddy retrieves course passages and never presents model memory as a citation.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

### Support and stretch

Provide one short approved passage; separate instruction, **input**, **output** and **evidence**. Stretch: introduce an unsupported claim or conflicting source and validate claim by claim with a fallback.

### Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

### Bridge to the next week

Next: Week 18: **Retrieval**-Augmented Generation. Retrieve today's **evidence** and identify what carries forward and what is new.

### Support and fresh reassessment

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

W4: Filter authorisation and active status before **evidence** reaches the student generation path: P only. Hiding a final citation does not undo exposed teacher content. Require intermediate reasoning and the stated scope.

Fresh retry: Student **corpus** has authorised A, restricted B and expired C. Select eligible **index**ed **evidence**.

Expected reasoning: A only; apply access and active-version filters before **retrieval** results enter the response context.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Next connection: Chunking and document structure. Preserve the current case record and retrieve its **evidence** boundary before moving on.

### Plan the delayed return

Suggested return points in this level: ai-3/18, ai-3/20, ai-3/23. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

| Record | Teacher evidence |
| --- | --- |
| Learner code and date | |
| Concept and case used; prior exposure | |
| Explanation / trace / limits | |
| Support and first repair | |
| Changed case and delayed outcome | |
| Next teaching action | |

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

### Project evidence from this case

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

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## Prepare, teach and assess

> Use this preparation summary alongside the detailed Model Memory vs Retrieval lesson; it is a checklist rather than a replacement for the full sequence.

### Teaching target

Teaching target: Explain the **mechanism** and terminology of **retrieval** and **index**ing

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
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> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
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>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: **Index** v1 contains notice N1 for room A. **Corpus** v2 supersedes it with N2 for room B, but the query still returns N1. What can be reported?

Show the intermediate steps and give the conclusion with its scope.

### Supported practice answer

Reasoned practice answer: Teacher explanation: The stale **index** returns old **evidence**. It cannot establish current room approval; refresh against authorised **corpus** v2 or report unresolved **freshness**.

Check every learner before discussion: Use the worked case for **Retrieval** and **index**ing. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

### Repair the misconception

Misconception repair: High **retrieval** similarity does not override permissions or make an obsolete source current.

### Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the

reasoning goal.

### Extend the reasoning

Extension: LIB-v = 2 is unavailable. Require a **prediction** and a reason before comparing. Expected result: Report that the current time cannot be verified.

Reason: The unsupported old guess does not become current **evidence** when **retrieval** fails.

### Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

| Criterion | Not yet | With support | Independent |
|---|---|---|---|
| Explanation | Names an answer without a deciding fact | Uses a hint to link fact and decision | Explains the deciding fact in own words |
| Trace or test | Cannot connect input and result | Completes steps with prompts | Traces or tests a changed case accurately |
| Limits and revision | Claims more than evidence supports | Names a limit after a prompt | States a defensible limit and useful next test |

### Project checkpoint

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

# Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

### For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and **index**ing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

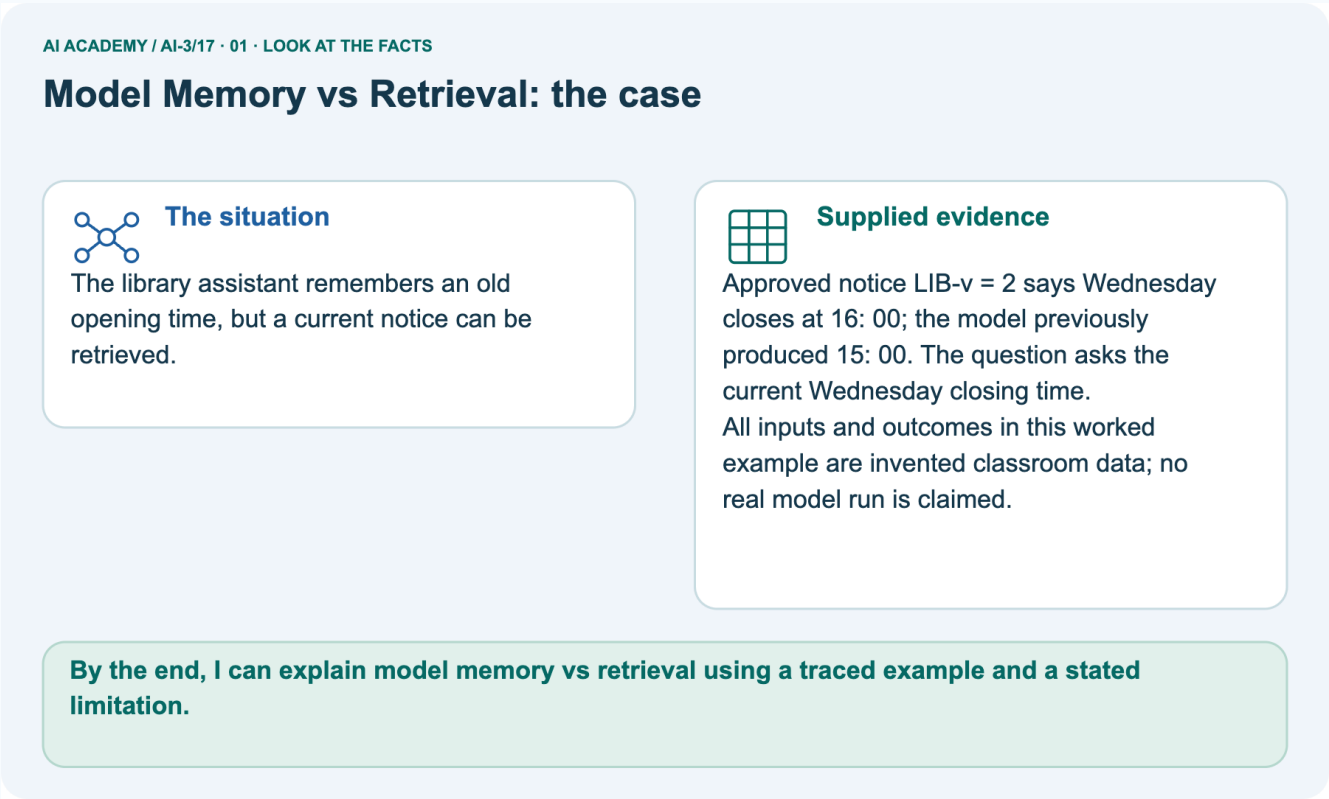
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

## Visual concept journey

> Show one Model Memory vs Retrieval graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

### 01. Model Memory vs Retrieval: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

The library assistant remembers an old opening time, but a current notice can be retrieved. Supplied facts: Approved notice LIB-v = 2 says Wednesday closes at 16: 00; the model previously produced 15: 00.

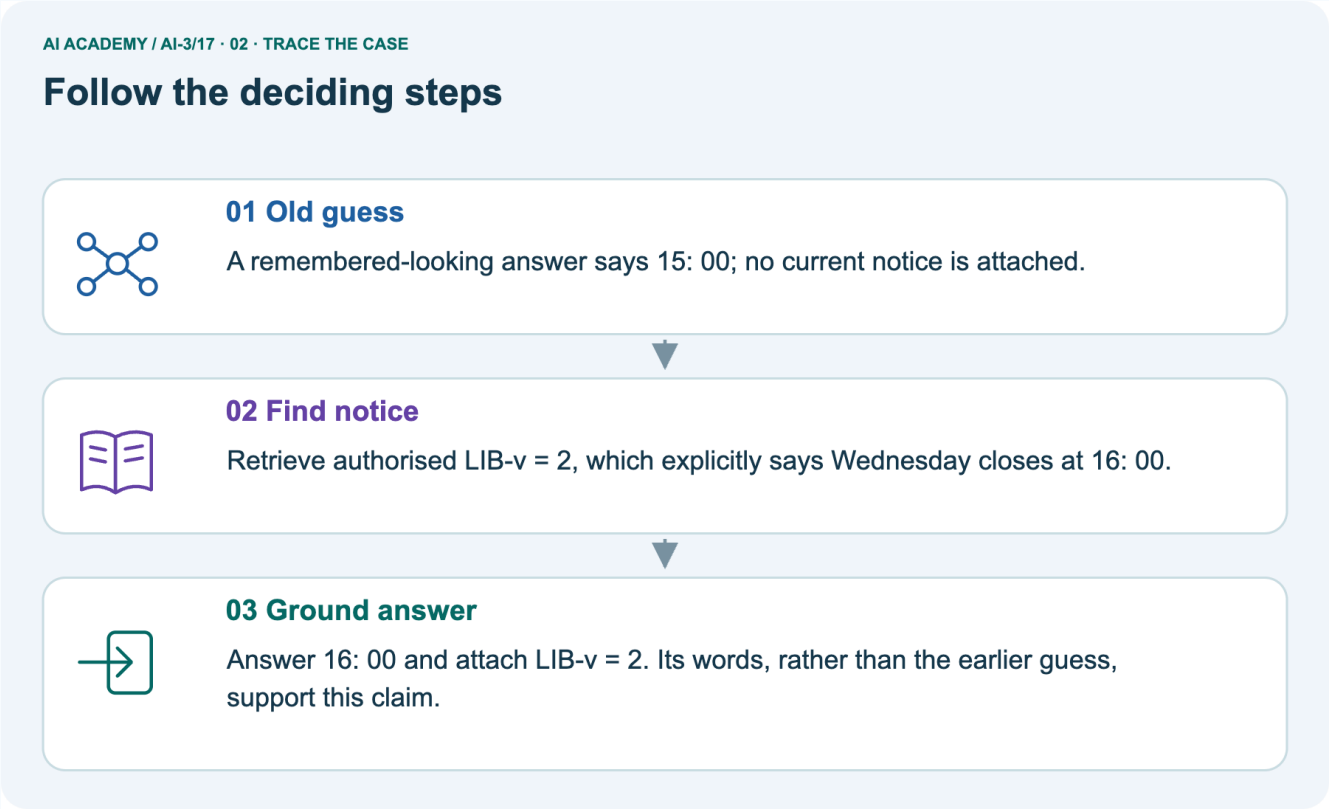
The question asks the current Wednesday closing time. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

### Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

### 02. Follow the deciding steps




Follow the labelled picture with the matching explanation. Open the vector diagram

Old guess: A remembered-looking answer says 15: 00; no current notice is attached. → Find notice: Retrieve authorised LIB-v = 2, which explicitly says Wednesday closes at 16: 00. → Ground answer: Answer 16: 00 and attach LIB-v = 2.

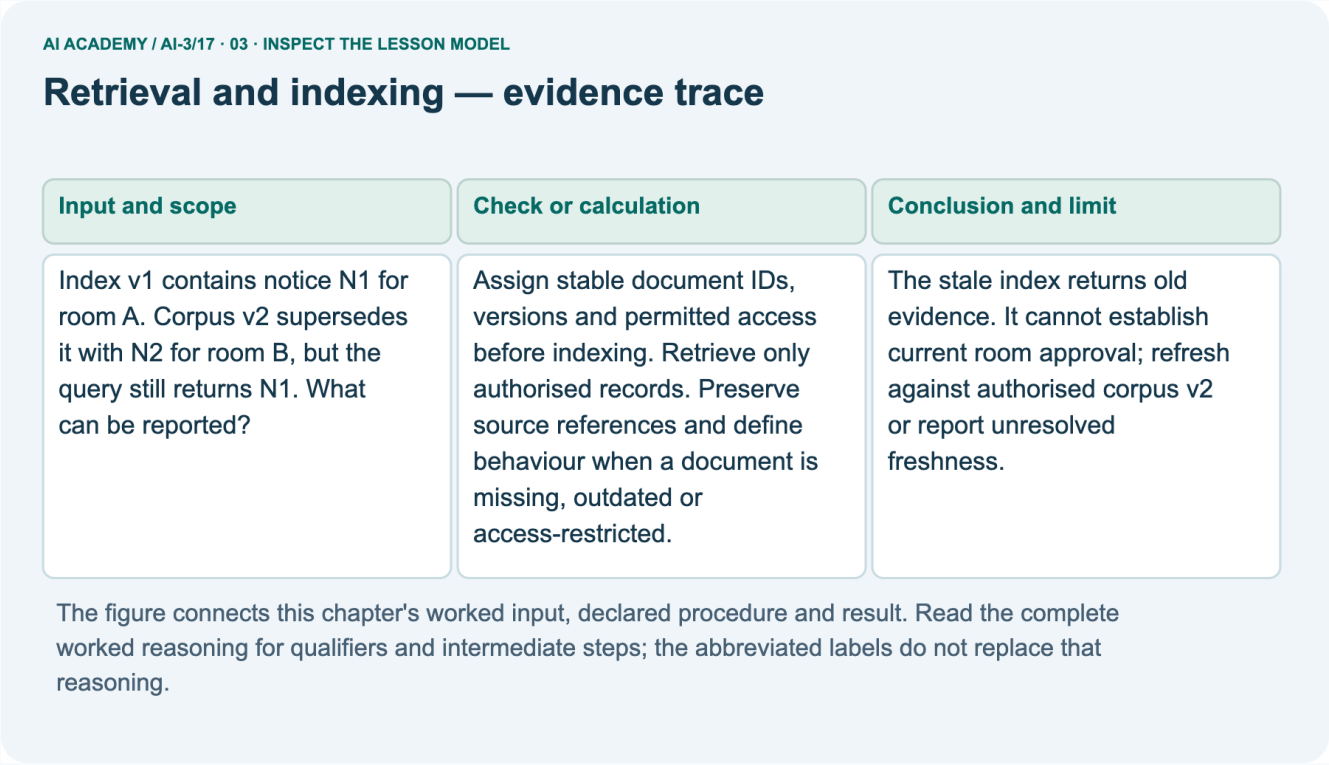
Its words, rather than the earlier guess, support this claim.

Point to each numbered stage and explain which recorded **input** it uses.

### Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

### 03. Retrieval and indexing — evidence trace




Follow the labelled picture with the matching explanation. Open the vector diagram

**Retrieval** and **index**ing — **evidence** trace. **Index** v1 contains notice N1 for room A. **Corpus** v2 supersedes it with N2 for room B, but the query still returns N1.

What can be reported? | Assign stable document IDs, versions and permitted access before **index**ing. Retrieve only authorised records.

Preserve source references and define behaviour when a document is missing, outdated or access-restricted. | The stale **index** returns old **evidence**.

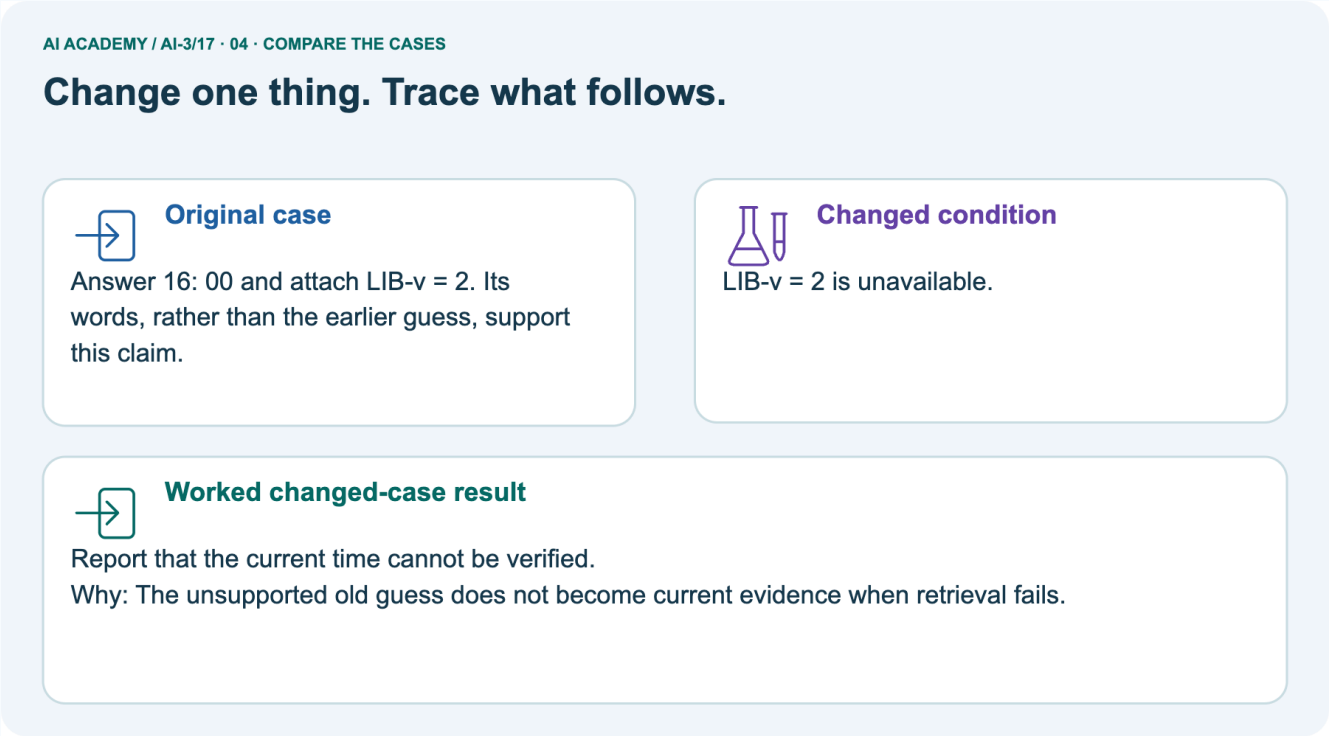
It cannot establish current room approval; refresh against authorised **corpus** v2 or report unresolved **freshness**.

Read the labels and values. Explain the relationship the model shows.

### Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Answer 16: 00 and attach LIB-v = 2. Its words, rather than the earlier guess, support this claim. Changed condition: LIB-v = 2 is unavailable. Expected result: Report that the current time cannot be verified.

Reason: The unsupported old guess does not become current **evidence** when **retrieval** fails.

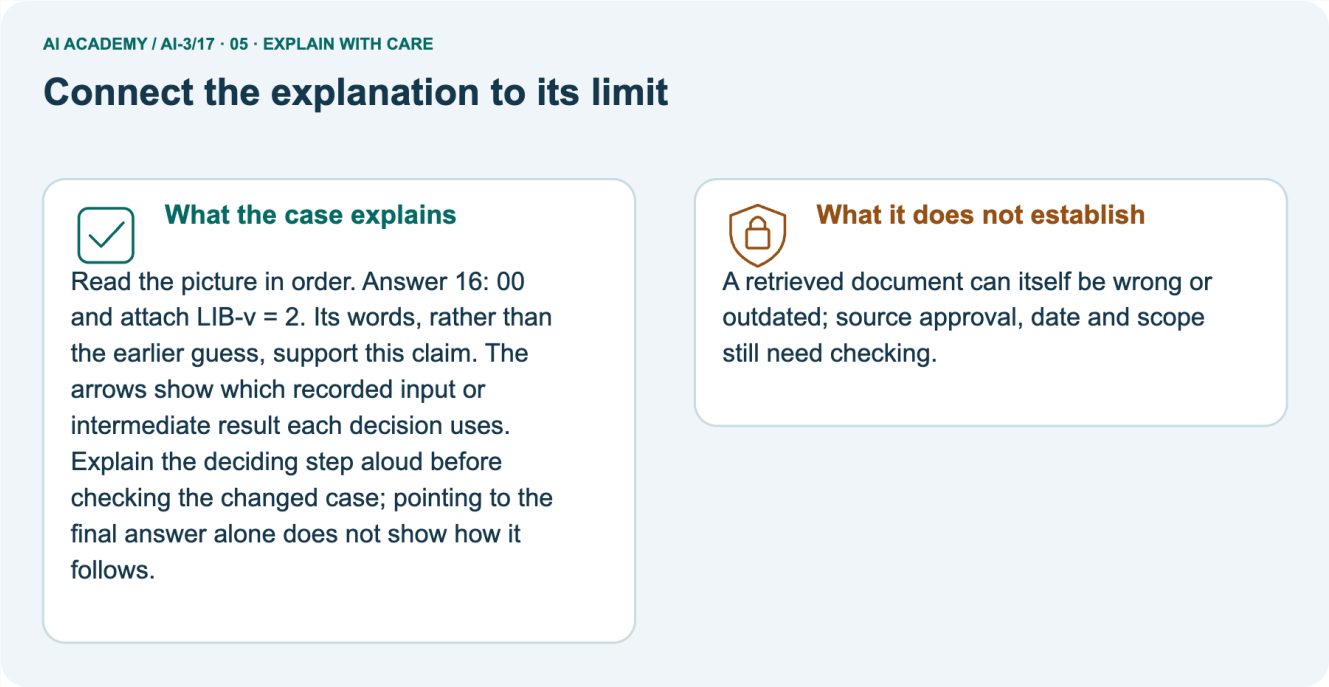
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

### Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The unsupported old guess does not become current **evidence** when **retrieval** fails.

### 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Answer 16: 00 and attach LIB-v = 2. Its words, rather than the earlier guess, support this claim. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A retrieved document can itself be wrong or outdated; source approval, date and scope still need checking.

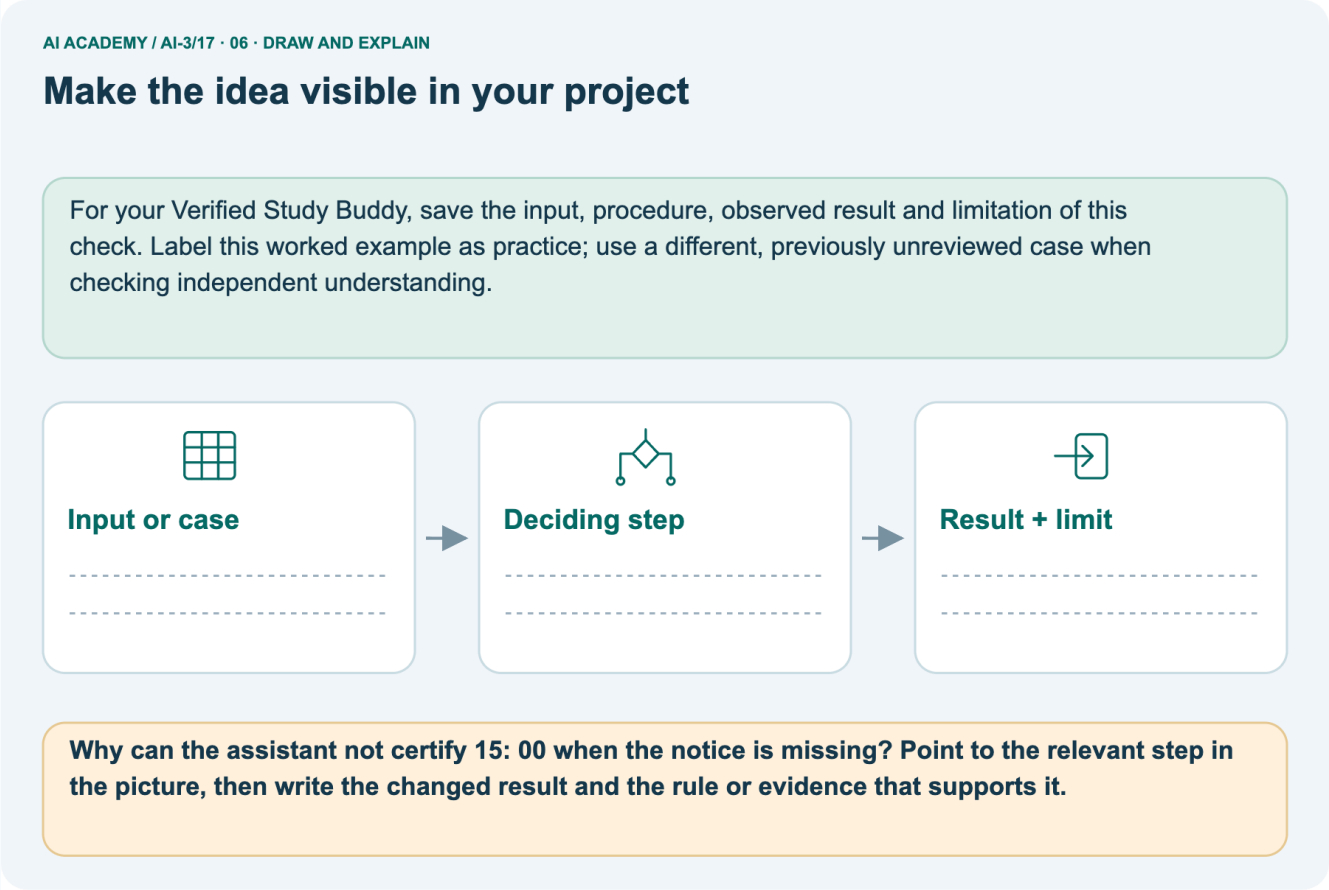
Explain the case, then point to a limit on the claim.

### Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A retrieved document can itself be wrong or outdated; source approval, date and scope still need checking.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why can the assistant not certify 15: 00 when the notice is missing? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

### Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: 15: 00 came from an unsupported earlier **output**. Missing current **evidence** requires an uncertainty statement, not substitution. Changed-case check: Report that the current time cannot be verified.

The unsupported old guess does not become current **evidence** when **retrieval** fails.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.